

Medium Level Conditional Statements (21-40)

21. Find the largest of three numbers.

```
In [3]: import math
num1 = int(input("Enter first number:"))
num2 = int(input("Enter second number:"))
num3 = int(input("Enter third number:"))

num4 = max (num1, num2, num3)
print(f"Among these {num1}, {num2} and {num3} numbers, the largest number is {num4}")
```

Among these 5, 7 and 9 numbers, the largest number is 9.

22. Determine if a number is a prime number.

```
In [7]: num = int(input("Enter a number: "))

# Prime numbers are greater than 1
if num > 1:
    for i in range(2, int(num**0.5) + 1): # check up to square root of num
        if num % i == 0:
            print(f"{num} is not a prime number")
            break
    else:
        print(f"{num} is a prime number")
else:
    print(f"{num} is not a prime number (prime numbers are greater than 1)")
```

99 is not a prime number

23. Check if a person is eligible for a driving license (age 18+, passed the driving test).

```
In [15]: age = int(input("How old are you?"))
test = str(input("Have you passed the driving test(Y/N)?"))

if age >= 18 and test=="Y":
    print("Congrat! You are eligible for driving license.")
else:
    print("Sorry, you are not eligible for driving license now.")
```

Congrat! You are eligible for driving license.

24. Write a program to determine whether a triangle is equilateral, isosceles, or scalene.

```
In [16]: x = int(input("Enter first side: "))
y = int(input("Enter second side: "))
z = int(input("Enter third side: "))

# First check if it is a valid triangle using triangle inequality
if (x + y > z) and (x + z > y) and (y + z > x):
    if x == y == z:
        print("The triangle is Equilateral.")
    elif x == y or y == z or x == z:
        print("The triangle is Isosceles.")
    else:
        print("The triangle is Scalene.")
else:
    print("The given sides do not form a valid triangle.")
```

The triangle is Isosceles.

25. Determine if a student passes or fails based on a passing mark of 40.

```
In [19]: marks = int(input("Enter your marks"))
if marks > 40:
    print("Congrat, you passed the test!")
else:
    print("Sorry, you did NOT pass the test!")
```

Congrat, you passed the test!

26. Check if a number is a palindrome (same forward and backward).

Firts option to check number only

```
In [28]: num = input("Enter a number:")

if num == num[::-1]:
    print(f"{num} is a palindrome!")
else:
    print(f"{num} is not a palindrome.")
```

121 is a palindrome!

Option two - to check both number and text

```
In [24]: text = input("Enter a word or number:")

# Convert input to lowercase (to ignore case sensitivity)
text = text.lower()

# Remove spaces (in case of phrases like 'nurses run')
text = text.replace(" ", "")

# Check if the string is equal to its reverse
if text == text[::-1]:
    print(f"'{text}' is a palindrome!")
else:
    print(f"'{text}' is not a palindrome.")
```

'madam' is a palindrome!

27. Calculate the electricity bill based on consumption: \$5 per unit for the first 100 units, \$10 per unit for the next 200, \$15 per unit for anything above 300 units.

```
In [35]: units = int(input("How many electricity units did you consume this month? "))

if units <= 100:
    bill = units * 5
elif units <= 300: # next 200 units (101-300)
    bill = 100 * 5 + (units - 100) * 10
else: # units above 300
    bill = 100 * 5 + 200 * 10 + (units - 300) * 15

print(f"Your electricity bill is: ${bill}")
```

Your electricity bill is: \$1000

28. Find the grade of a student based on marks (90+ A, 80-89 B, etc.).

```
In [48]: marks = int(input("Enter your marks:"))

if marks >= 90:
    print("Grade A")
elif marks >= 80:
```

```
    print ("Grade B")
elif marks >= 40:
    print ("Grade C")
else:
    print ("Sorry, you did not pass the test, work hard for next time!")
```

Grade A

29. Determine if a given date is valid (considering month length and leap year for February).

to be continued.....

In []:

In []:

In []: